

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA51 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO3: *Examine cryptographic hash algorithms, digital signature schemes, and assess Kerberos and X.509 authentication frameworks for ensuring integrity, authentication, and non-repudiation.CO (BL4 , BL5)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A CASE STUDY

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

1. Weak Password Hashing in Web Application

A web application stores user passwords using unsalted SHA-1 hashes. After a breach, attackers quickly crack multiple passwords using precomputed tables.

Analyze how the lack of salting and use of SHA-1 contributed to this vulnerability.

Recommend a secure password hashing strategy and justify your choice.

2. Digital Signature Compromise in Financial System

A financial institution uses digital signatures for transaction approval. An attacker gains access to a user's private key and performs unauthorized transactions.

Evaluate how the compromise affects authentication and non-repudiation. Propose mitigation strategies to secure private keys and prevent such incidents.

3. Kerberos Authentication Failure in Distributed System

A distributed system using Kerberos experiences frequent authentication failures due to time synchronization issues between servers.

Analyze how time synchronization impacts Kerberos functionality. Suggest solutions to ensure reliable authentication and maintain system security.

4. Certificate Authority Breach in PKI System

A Certificate Authority (CA) is compromised, leading to the issuance of fake X.509 certificates for trusted domains.

Evaluate the impact of this breach on system trust and secure communication. Propose mechanisms to detect and mitigate such attacks in a PKI environment.

5. Integrity Verification in Cloud Data Storage

A cloud service provider uses hashing to verify file integrity, but users report data tampering incidents. Investigation reveals improper hash validation practices.

Analyze how incorrect implementation of hashing can lead to integrity failures.

Recommend a robust integrity verification mechanism and justify its effectiveness.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand